

From maths to machines

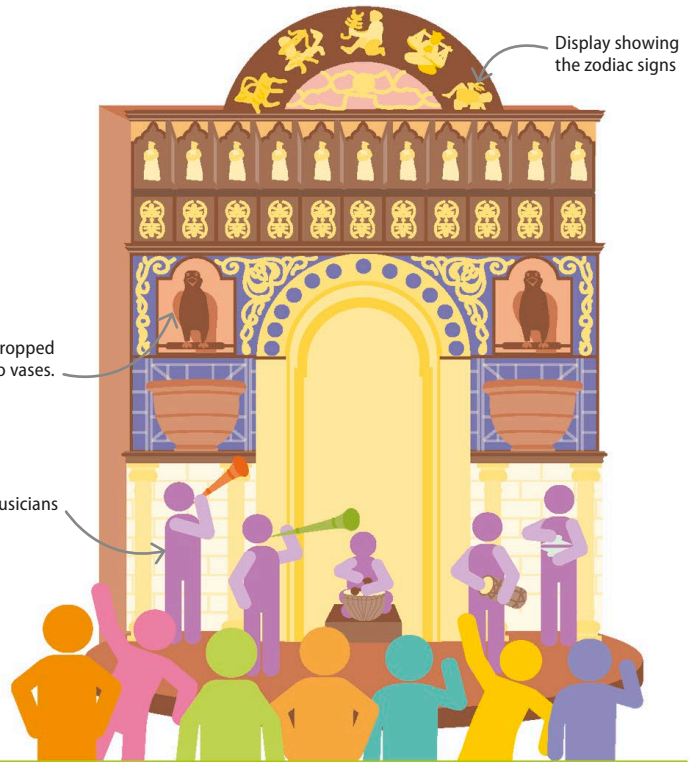
From the 8th to the 14th centuries, a lot of mathematical work that ultimately proved crucial to computing was done in the Islamic world. Scholars translated mathematical texts by ancient Greek and Indian mathematicians into Arabic, and built on the knowledge they contained to develop new methods of calculation. Two particularly notable mathematicians were Muhammad ibn Musa al-Khwarizmi (c.780–850), from whose name the word algorithm comes, and Al-Kindi (c.801–873), who developed techniques still used in cryptography today.

▷ Al-Jazari castle water clock

A 12th-century engineer, Ismail al-Jazari (1135–1206) invented many ingenious mechanical devices. His castle clock was particularly complex and could be programmed to take into account the different lengths of day and night across the year.

Falcon automatons dropped balls into vases.

Mechanical musicians



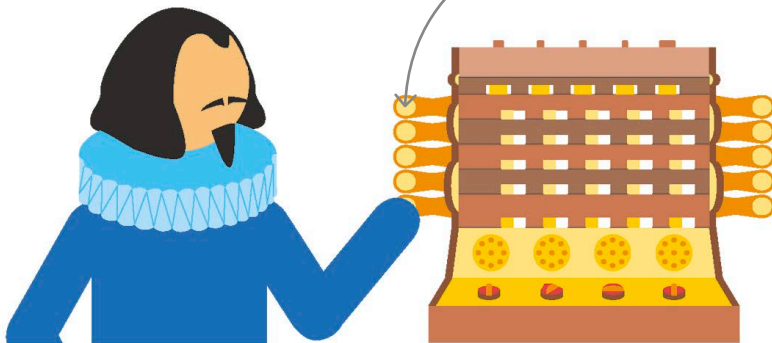
Napier and Schickard

Scottish mathematician John Napier (1550–1617) created a manually operated calculating device called Napier's bones. A set of square rods carved from bone inscribed with numbers, it made multiplication, division, and finding square roots much easier. Napier based his device on an Arabic method introduced to Europe by the Italian mathematician Fibonacci (1175–1250).

▽ Schickard's calculating clock

German astronomer William Schickard built a calculating machine by remodelling Napier's bones as cylinders. His clock could add and subtract six-digit numbers.

Calculation results were read through windows in these wooden slats.



BIOGRAPHY

Ada Lovelace

English mathematician Ada Lovelace (1815–1852) created the world's first computer program when she wrote the earliest algorithm to be processed on English scientist Charles Babbage's (1791–1871) Analytical Engine machine in 1844. She was the first to see that computers could do so much more than basic number-crunching and calculations.

